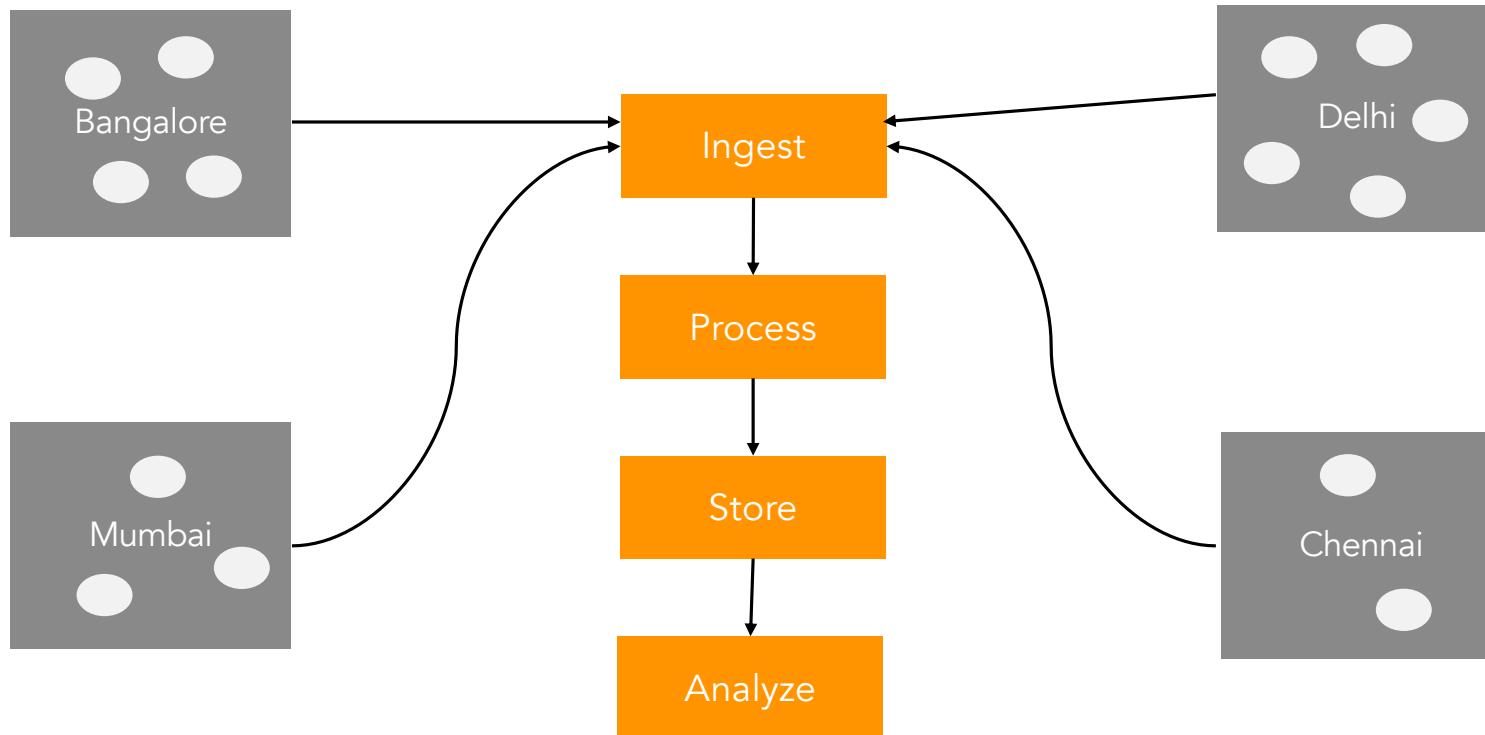


Air Quality Index (AQI) Analysis

www.3aayaam.in

Data Collection



Schema

Sensor Data

SensorID
SensorStationID
ModelNumbe
Manufacturer
TargetPollutant
CalibrationDate
FirmwareVersion

Station Data

StationID
StationName
Address
City
State
Country
InstallationDate

Pollutant Data

PollutantCode
PollutantName
PollutantUnit
DangerousThreshold
HealthyThreshold

Stream Data

EventID
EventTimeIsoFormat
SensorID
SensorStationID
CO2Level
PM25Level
PM10Level
COLevel
NO2Level
O3Level
SO2Level
WeatherTemperature
WeatherHumidity

Air Quality Index

- Measures the quality of breathable air.
- For every pollutant, the index is calculated as below:

$$I_p = \frac{I_{Hi} - I_{Lo}}{BP_{Hi} - BP_{Lo}} \times (C_p - BP_{Lo}) + I_{Lo}$$

- AQI is the highest of all the indexes.
- C_p is the value received from Sensor.
- (I_{Hi}, I_{Lo}) and (BP_{Hi}, BP_{Lo}) values can be found in the table shown...

Air Quality Index

$$I_p = \frac{I_{Hi} - I_{Lo}}{BP_{Hi} - BP_{Lo}} \times (C_p - BP_{Lo}) + I_{Lo}$$

AQI Category	AQI Range (I_{Lo} – I_{Hi})	PM2.5 ($\mu g/m^3$)	PM10 ($\mu g/m^3$)	CO (ppm)	NO2 (ppb)
Good	0 – 50	0.0 – 9.0	0 – 54	0.0 – 4.4	0 – 53
Moderate	51 – 100	9.1 – 35.4	55 – 154	4.5 – 9.4	54 – 100
Unhealthy (Sens.)	101 – 150	35.5 – 55.4	155 – 254	9.5 – 12.4	101 – 360
Unhealthy	151 – 200	55.5 – 125.4	255 – 354	12.5 – 15.4	361 – 649
Very Unhealthy	201 – 300	125.5 – 225.4	355 – 424	15.5 – 30.4	650 – 1249

Air Quality Index

$$I_p = \frac{I_{Hi} - I_{Lo}}{BP_{Hi} - BP_{Lo}} \times (C_p - BP_{Lo}) + I_{Lo}$$

EventID,EventTimeIsoFormat,SensorID,SensorStationID,CO2Level,PM25Level,PM10Level,COLevel,NO2Level,O3Level,SO2Level,WeatherTemperature,WeatherHumidity

8c1bxxx,2026-01-24T19:39:43,cbe1xxxx,53554839,721.16,52.65,140.73,5.97,127.53,0.054,165.312,11.1260,46.417

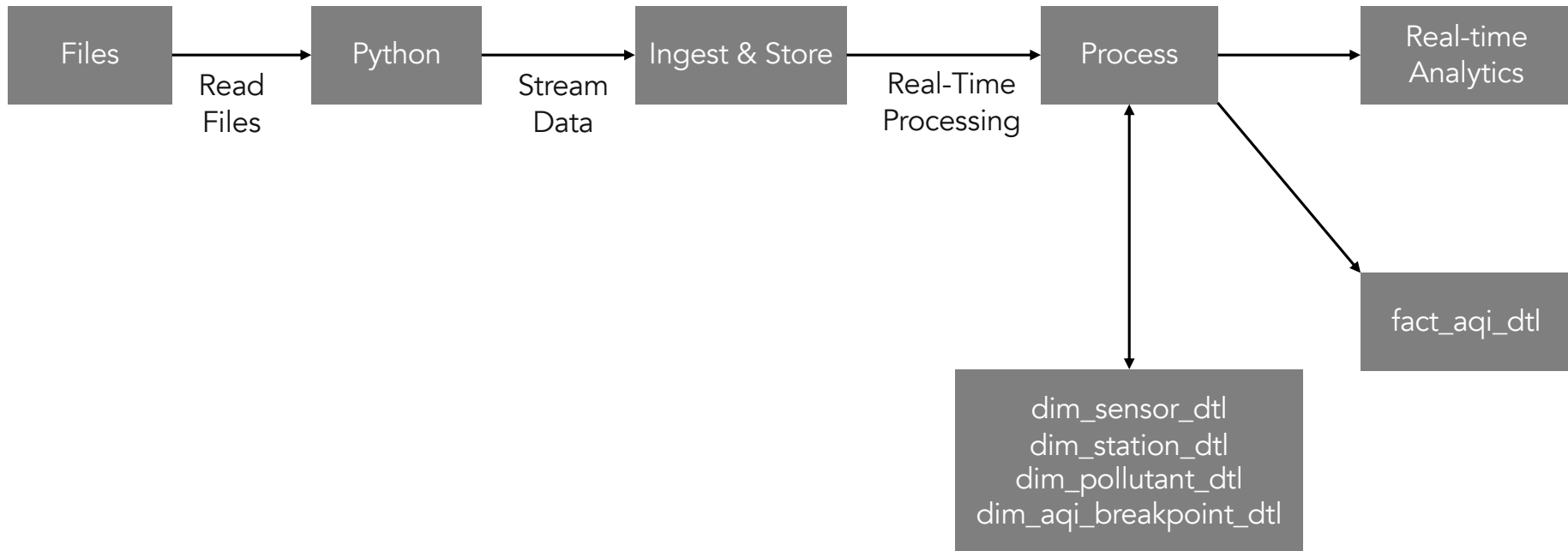
PM25 Level – 52.65.

$$I_p = (150 - 101)/(55.4 - 35.5) * (52.65 - 35.5) + 101$$

$$= (49/19.9) * 17.15 + 101 = 143.23$$

AQI Category	AQI Range (I_{Lo} – I_{Hi})	PM2.5 ($\mu g/m^3$)	PM10 ($\mu g/m^3$)	CO (ppm)	NO2 (ppb)
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Stream & Batch Processing



Business Processed

- Whenever the AQI for the last 15 mins (of any pollutant) falls in “unhealthy” category, store in real-time storage.
- The raw data should be stored as is in a separate table.
- Record the time when temperature is more than 30 and humidity is more than 65 for every 15 mins.
- Store the avg. AQI for every 15 mins in the fact table.
- Store avg. temperature and humidity for every 15 mins in the fact table.